Mechanistic Interpretability via Riemannian Information Geometry on Activation Space

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Yes—with one crucial geometric extension: Information Geometry naturally introduces **dual connections** (∇, ∇^*) in addition to the standard Riemannian Levi-Civita connection.

When you treat a layer's activation space $H \subseteq \mathbb{R}^d$ as a manifold equipped with the **pullback Fisher Information Metric** from the network's output distribution $p(y \mid h)$, the core criteria of **Mechanistic Interpretability** (linear representations, disentanglement, superposition, and causal abstraction) translate directly into exact geometric conditions on the metric $g_{ij}(h)$ and the **Christoffel symbols** $\Gamma^k_{ij}(h)$.

Below is the dictionary translating Mechanistic Interpretability criteria into Riemannian Information Geometry on activation space.

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1 The Setup: The Metric on Activation Space

Let $h \in \mathbb{R}^d$ be the hidden activations at layer ℓ, and let the rest of the network define a conditional probability distribution over outputs $p(y \mid h)$ (e.g., via a softmax over logits $z(h)$).

The **Fisher Information Metric (FIM)** pulled back to activation space h is:

$$g_{ij}(h) = \mathbb{E}_{y \sim p(\cdot|h)} \left[\frac{\partial \log p(y \mid h)}{\partial h^i} \frac{\partial \log p(y \mid h)}{\partial h^j} \right] = J(h)^\top I_z(h) J(h)$$

where $J(h) = \frac{\partial z}{\partial h}$ is the Jacobian of the downstream network and $I_z(h)$ is the covariance of the output distribution.

- **What $g_{ij}(h)$ measures:** It measures **causal sensitivity**. A small step δh has squared information distance:

$$ds^2 = g_{ij}(h)\, \delta h^i\, \delta h^j = 2\, D_{\mathrm{KL}}(p(\cdot \mid h) \parallel p(\cdot \mid h + \delta h))\,.$$

  Directions in the nullspace of $g_{ij}(h)$ have *zero downstream effect* on the output.

2 Translating Mechanistic Interpretability Criteria into Geometry

3 Why Christoffel Symbols Are the Exact Measure of "Non-Linearity of Meaning"

Returning to the original question: **Why do we care that Γ^k_{ij} is coordinate-dependent?**

Mechanistic Interpretability Criterion	**Riemannian / Information Geometric Definition**	**Condition on g_{ij} and Γ^k_{ij}**
1. The Linear Representation Hypothesis *(Features are constant directions)*	**Autoparallelism / Flatness:** Moving along a feature direction transports the feature meaning without rotating it into other features.	$\Gamma^k_{ij}(h) = 0$ in the feature coordinate basis (it forms an **affine / Riemann normal** coordinate system).
2. Feature Disentanglement & Monosemanticity *(Features don't interfere)*	**Metric Orthogonality & Decomposability:** The manifold splits into a Riemannian product of independent submanifolds $M = M_1 \times M_2 \times \dots$	$g_{ij}(h)$ **is block-diagonal** ($g_{ij} = 0$ for $i \neq j$) and $\partial_m g_{ij} = 0$ across blocks (Γ^k_{ij} doesn't mix blocks).
3. Polysemanticity & Superposition *(More features F than neurons d)*	**Degenerate / Frame-Bundle Packing:** $F > d$ vector fields $\{v_a\}_{a=1}^F$ packed into tangent space T_hM with minimal information interference.	Off-diagonal metric inner products $g_h(v_a, v_b) = v_a^i g_{ij}(h) v_b^j \approx \epsilon$ are kept small on the active data submanifold.
4. Contextual Routing / Attention Gating *(Feature A changes how Feature B is read)*	**Non-zero Christoffel Cross-Terms / Holonomy:** Moving along feature i rotates or scales the downstream sensitivity of feature j.	$\Gamma^k_{ij}(h) \neq 0$ for $i \neq j$, and the **Riemann curvature** $R^l_{ijk} \neq 0$ measures non-commutativity of features.
5. Causal Scrubbing / Activation Patching *(Replacing h along a subspace)*	**Geodesic Projection (Information Bottleneck):** Projecting h onto a submanifold S that minimizes KL-divergence to the true output.	Orthogonal projection with respect to g_{ij} (specifically, ∇^***-geodesic projection** in dually flat geometry).

In Mechanistic Interpretability, **choosing a dictionary of features (e.g., via a Sparse Autoencoder or coordinate change $h \mapsto f(h)$) IS choosing a coordinate system on activation space.**

- Suppose you are in the raw neuron basis $h^1, \dots, h^d$. Because neurons are polysemantic and downstream layers are nonlinear, a concept vector $v(h)$ ("is a Golden Gate Bridge") changes its direction in neuron space depending on context h. Thus, in neuron coordinates, $\Gamma^k_{ij} \neq 0$.
- **The goal of finding an "interpretable feature basis"** $f^a(h)$ is literally the geometric problem of finding a coordinate transformation $h \mapsto f(h)$ that **flattens the connection**—making $\Gamma^{c'}_{a'b'} \approx 0$ so that "straight lines in feature space correspond to constant causal paths in output distribution space."
- **When is the Linear Representation Hypothesis strictly impossible?** Because the **Riemann Curvature Tensor** R^l_{ijk} *is* coordinate-free, if $R^l_{ijk}(h) \neq 0$ on the activation manifold, **no coordinate transformation (not even a non-linear SAE) can make** $\Gamma^k_{ij} = 0$ **everywhere.** Non-zero Riemann curvature is the exact, coordinate-free obstruction to globally linear, non-interacting features!

4 The Information Geometry Secret Weapon: Amari's Dual Connections ($\nabla^{(e)}$ and $\nabla^{(m)}$)

Pure Riemannian geometry only uses the Levi-Civita connection $\nabla^{(0)}$ (derived from g_{ij}). **Information Geometry** (pioneered by Shun-ichi Amari) goes one step further and defines a pair of **dual connections**:

1. **The e-connection ($\nabla^{(e)}$, exponential / log-linear):** A straight line ($\Gamma_{ij}^{(e)k} = 0$) in this geometry means **logits add linearly**.
    - *Crucial insight:* If the final layer of a Transformer is just a linear unembedding + Softmax ($p(y \mid h) \propto \exp(Wh)$), then the final hidden layer h is **naturally e-flat ($\Gamma_{ij}^{(e)k} = 0$)**! This is the mathematical reason why the **Linear Representation Hypothesis works so well in the final residual stream** of LLMs: in the e-connection of Information Geometry, linear directions in h *are* exact geodesics of log-probability!
2. **The m-connection ($\nabla^{(m)}$, mixture / expectation):** A straight line ($\Gamma_{ij}^{(m)k} = 0$) in this geometry means **probabilities mix linearly** (expectation parameters interpolate linearly).

When an intermediate layer h_ℓ passes through nonlinear MLPs and Attention layers before reaching the logits, it embeds a curved submanifold inside that e-flat output space. The **second fundamental form (extrinsic curvature)** and induced **Christoffel symbols** $\Gamma_{ij}^{(e)k}$ of layer ℓ measure *precisely* how much the remaining layers bend the feature representations before readout.

5 Two Important Caveats to Keep in Mind

If you want to formalize mechanistic interpretability this way, there are two mathematical hurdles you have to handle:

1. **The Metric is Singular (Rank-Deficient):** If activation space has dimension $d = 4096$ and the output classification has $C = 10$ classes (or even $V = 50,000$ tokens where only a few tokens have non-zero probability at once), the Fisher metric $g_{ij}(h)$ is **degenerate** (rank $\leq \min(d, V-1)$, and effectively much lower). Because g_{ij} is not invertible, g^{km} does not exist everywhere!
    - *Solution:* You either regularize the metric ($g_{ij} + \lambda\delta_{ij}$, moving to a Fisher-Rao + Euclidean geometry on the active data manifold) or work on the **quotient manifold** modulo the nullspace $\ker(g)$ (which neatly quotients out causally irrelevant activation directions!).
2. **ReLU / GELU Smoothness:** ReLU networks create piecewise-linear manifolds where derivatives jump at activation boundaries (concentrating all curvature onto codimension-1 polytope boundaries), whereas GELU/SiLU (used in modern Transformers) make the activation manifold genuinely C^∞ smooth, so Γ_{ij}^k is well-defined everywhere.